IPython: researchcodes/BuildI

```
sage: run test_equations.py
************************************************************************************
************ The locus polynomials that can be universally used  **************
************************************************************************************

------------------------------------------------------------
# !!! Locus polynomial for cyclic subgroups of prime order 5 !!! #
------------------------------------------------------------

5*T^6 + 38*a*0*T^5 + (55*a^2*0^2 + 420*b*0 + 100*a)*T^4 + (-60*a^3*0^3 - 640*b^
2*0^3 + 560*a*b*0^2 - 240*a^2*0 + 800*b)*T^3 + (35*a^4*0^4 + 160*a*b^2*0^4 + 28
0*a^2*b*0^3 - 200*a^3*0^2 + 720*b^2*0^2 - 320*a*b*0 - 400*a^2)*T^2 + (-10*a^5*0
^5 - 80*a^2*b^2*0^5 - 80*a^3*b*0^4 + 80*a^4*0^3 - 960*b^3*0^4 + 1280*a*b^2*0^3
- 480*a^2*b*0^2 + 160*a^3*0 + 1200*b^2*0 - 640*a*b)*T + a^6*0^6 - 32*a^3*b^2*0^
6 + 100*a^4*b*0^5 - 256*b^4*0^6 - 60*a^5*0^4 + 640*a*b^3*0^5 - 320*a^2*b^2*0^4
- 80*a^4*0^2 + 640*b^3*0^3 - 800*a*b^2*0^2 - 64*a^2*b*0 - 400*b^2

The corresponding normalized polynomial is:
T^6 + 38/5*a*0*T^5 + (11*a^2*0^2 + 84*b*0 + 20*a)*T^4 + (-12*a^3*0^3 - 128*b^2*
0^3 + 112*a*b*0^2 - 48*a^2*0 + 160*b)*T^3 + (7*a^4*0^4 + 32*a*b^2*0^4 + 56*a^2*
b*0^3 - 40*a^3*0^2 + 144*b^2*0^2 - 64*a*b*0 - 80*a^2)*T^2 + (-2*a^5*0^5 - 16*a^
2*b^2*0^5 - 16*a^3*b*0^4 + 16*a^4*0^3 - 192*b^3*0^4 + 256*a*b^2*0^3 - 96*a^2*b*
0^2 + 32*a^3*0 + 240*b^2*0 - 128*a*b)*T + 1/5*a^6*0^6 - 32/5*a^3*b^2*0^6 + 20*a
^4*b*0^5 - 256/5*b^4*0^6 - 12*a^5*0^4 + 128*a*b^3*0^5 - 64*a^2*b^2*0^4 - 16*a^4
*0^2 + 128*b^3*0^3 - 160*a*b^2*0^2 - 64/5*a^2*b*0 - 80*b^2

------------------------------------------------------------
# !!! Locus polynomial for cyclic subgroups of prime order 3 !!! #
------------------------------------------------------------
It corresponds to the normalization of the 3-division polynomial:
T^4 + 2*a*T^2 + 4*b*T - 1/3*a^2

************************************************************************************
****** Working with a field whose characteristic has a bit length ~ 128  ******
************************************************************************************

# !!! The parameters used in the construction for subgroups of order 3 !!! #
------------------------------------------------------------------------

The domain E:
Elliptic Curve defined by y^2 = x^3 + 151593404684078088939912405041019744186*x
 + 110720351286387880119917917146075950116 over Finite Field of size 2272791501
589473372194761956248269889 89

The kernel coefficients:
[1, 159347995054138985157208878784556237851]
```

```
The embedding field element:
1

The subgroup identifier:
679311551048083520622673168402707511 38

The isogeny codomain:
Elliptic Curve defined by y^2 = x^3 + 17401496757886172415245943897276869985 2*x
 + 30143242131913694591194131428995088265 over Finite Field of size 22727915015
894733721947619562482698898 9

# !!! The parameters used in the construction for subgroups of order 5 !!! #
--------------------------------------------------------------------------

The domain E:
Elliptic Curve defined by y^2 = x^3 + 12463781619983409870567239960702001875 5*x
 + 13503258591522938579706381325046046013 7 over Finite Field of size 2272791501
589473372194761956248269889 89

The kernel coefficients:
[1, 19583842564485572336626202455921441082 1, 99558274962145220910018325496472973
3473]

The embedding field element:
θ

The subgroup identifier:
995582749621452209100183254964729734 73*θ + 31440724514091613853214171065612578168

The isogeny codomain:
Elliptic Curve defined by y^2 = x^3 + 19272792006463076977428834098972992774 6*x
 + 46181498767558708096100941927608420314 over Finite Field of size 22727915015
894733721947619562482698898 9


*************************************************************************
********** Testing if the locus polynomial satisfies the parameters  **********
*************************************************************************

The evaluation of Psi_5 at the subgroup identifier lambda is:
0
As expected the evaluation at the unique identfier is equal to zero!!!
sage:
```